

Demostració

Siguin z_1, z_2 les arrels del polinomi

$$x^2 + ax + b = 0, \quad a, b \in \mathbb{R}.$$

Com que els coeficients són reals, si les arrels no són reals, són conjugades:

$$z_2 = \overline{z_1}.$$

Per a qualsevol enter positiu n :

$$z_1^n + z_2^n = z_1^n + \overline{z_1}^n.$$

Com que

$$\overline{z_1^n} = \overline{z_1}^n,$$

tenim:

$$z_1^n + z_2^n = z_1^n + \overline{z_1^n} = 2 \operatorname{Re}(z_1^n).$$

Per tant,

$$z_1^n + z_2^n \in \mathbb{R}.$$

Cas concret: $x^2 - 2x + 2 = 0$

Resolem:

$$x = \frac{2 \pm \sqrt{4 - 8}}{2} = \frac{2 \pm \sqrt{-4}}{2} = 1 \pm i.$$

Així,

$$z_1 = 1 + i, \quad z_2 = 1 - i.$$

Forma polar

$$|z_1| = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \arg(z_1) = \frac{\pi}{4}.$$

$$z_1 = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right),$$

$$z_2 = \sqrt{2} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right).$$

Potències

$$z_1^n = (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right),$$

$$z_2^n = (\sqrt{2})^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right).$$

Sumant:

$$z_1^n + z_2^n = 2(\sqrt{2})^n \cos \left(\frac{n\pi}{4} \right).$$

Resultat final

$$z_1^n + z_2^n = 2(\sqrt{2})^n \cos\left(\frac{n\pi}{4}\right)$$